

Problem 18.14

John is receiving annual payments from a perpetuity-immediate of \$12,000. Krista is receiving annual payments from a perpetuity-due of \$10,000. If the present value of each perpetuity is equal, calculate i .

Solution

John's perpetuity is a perpetuity-immediate, so its present value is

$$12000a_{\infty} = 12000 \left(\frac{1}{i} \right).$$

Krista's perpetuity is a perpetuity-due, so its present value is

$$10000\ddot{a}_{\infty} = 10000 \left(\frac{1}{d} \right).$$

Since

$$d = \frac{i}{1+i},$$

we have

$$\frac{1}{d} = \frac{1+i}{i}.$$

Thus, Krista's present value is

$$10000 \left(\frac{1+i}{i} \right).$$

The present values are equal, so

$$12000 \left(\frac{1}{i} \right) = 10000 \left(\frac{1+i}{i} \right).$$

Multiplying both sides by i gives

$$12000 = 10000(1+i).$$

Therefore,

$$1+i = 1.2.$$

So

$$i = 0.2.$$

Hence,

$$\boxed{i = 20\%}.$$